

Differential Equations and Applications



Lecture 3: MATH 203i

Instructor: Dr. Sameh Dakroury

samohamed@nu.edu.eg

Integrating Factors

Example: Solve $(2y + 4x^3)dx + xdy = 0$

Not Separation.... Not Homog.... Not Reduced...

Check Exact : $\frac{\partial}{\partial y}(2y + 4x^3) = 2$ $\frac{\partial}{\partial x}(x) = 1$ ☹

.....Let us multiply the DE by x ??

$$(2xy + 4x^4)dx + x^2dy = 0$$

Check Exact : $\frac{\partial}{\partial y}(2xy + 4x^4) = 2x$ $\frac{\partial}{\partial x}(x^2) = 2x$

EXACT DE ☺ !!!

Integrating Factors

$$M(x, y)dx + N(x, y)dy = 0$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Then, we look for a function $\mu(x, y)$ such that if we multiply the DE by that function the new DE reduces to Exact DE. Following:

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0$$

$$\frac{\partial}{\partial y} (\mu(x, y)M(x, y)) = \frac{\partial}{\partial x} (\mu(x, y)N(x, y))$$

$$\frac{\partial}{\partial y} (\mu(x, y)M(x, y)) = \frac{\partial}{\partial x} (\mu(x, y)N(x, y))$$

$$\begin{aligned} & \mu(x, y) \frac{\partial M(x, y)}{\partial y} + M(x, y) \frac{\partial \mu(x, y)}{\partial y} \\ &= \mu(x, y) \frac{\partial N(x, y)}{\partial x} + N(x, y) \frac{\partial \mu(x, y)}{\partial x} \end{aligned}$$

After substitution, the evaluation of the function $\mu(x, y)$ will not be so easy. So, we will discuss two special **cases**:

- $\mu(x, y) = \mu(x)$, independent of y
- $\mu(x, y) = \mu(y)$, independent of x

Case of $\mu(x, y) = \mu(x)$:

Then, $\mu(x) \frac{\partial M(x, y)}{\partial y} + M(x, y) \frac{\partial \mu(x)}{\partial y} = \mu(x) \frac{\partial N(x, y)}{\partial x} + N(x, y) \frac{d\mu(x)}{dx}$

This formula can be simplified into:-

$$\frac{d\mu(x)}{\mu(x)} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} dx$$

The assumption μ is a function of x only is valid if the RHS is a function of x only, and in this case:

$$\mu(x) = e^{\int \left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \right) dx}$$

Case of $\mu(x, y) = \mu(y)$:

Then, $\mu(y) \frac{\partial M(x, y)}{\partial y} + M(x, y) \frac{d\mu(y)}{dy} = \mu(y) \frac{\partial N(x, y)}{\partial x}$

This formula can be simplified into:-

$$\frac{d\mu(y)}{\mu(y)} = - \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} dy$$

Also, the assumption μ is a function of y only is valid if the RHS is a function of y only, and in this case:

$$\mu(y) = e^{-\int \left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} \right) dy}$$

Summary:- Integrating Factor

$$M(x, y)dx + N(x, y)dy = 0$$

Case $\left(\frac{M_y - N_x}{N}\right)$ is a fun. of **only x** : $\mu(x) = e^{\int \left(\frac{M_y - N_x}{N}\right) dx}$

Case $\left(\frac{M_y - N_x}{M}\right)$ is a fun. of **only y** : $\mu(y) = e^{-\int \left(\frac{M_y - N_x}{M}\right) dy}$

Examples:-

$$(10 - 6y + e^{-3x})dx - 2dy = 0$$

$$xydx + (2x^2 + 3y^2 - 20)dy = 0$$

Example: Solve the following DE

$$ydx + (3x - xy + 2)dy = 0$$

$$\frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = 3 - y$$

This DE is not Exact.

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = y - 2$$

$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M}$ is a function of only y .

Then, μ is function of y : $\mu = \mu(y)$

$$\mu(y) = e^{-\int \left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} \right) dy} = e^{-\int \frac{y-2}{y} dy} = e^{-y+2\ln y} = y^2 e^{-y}$$

Next step is to multiply the DE by $\mu(y)$ and check that the DE reduces to Exact.

$$y^3 e^{-y} dx + e^{-y} (3xy^2 - xy^3 + 2y^2) dy = 0$$

$$M(x, y) = y^3 e^{-y}, \quad N(x, y) = e^{-y} (3xy^2 - xy^3 + 2y^2) \frac{\partial M}{\partial y}$$

$$= (3y^2 - y^3) e^{-y} \quad \frac{\partial N}{\partial x} = e^{-y} (3y^2 - y^3)$$

Then the new DE is Exact. Check that the general solution is

$$xy^3 e^{-y} - (2y^2 + 4y + 4) e^{-y} = c$$

Example:

$$(2y + 4x^3)dx + xdy = 0$$

$$M_y = \frac{\partial}{\partial y}(2y + 4x^3) = 2 \quad N_x = \frac{\partial}{\partial x}(x) = 1$$

$$\mu(x) = e^{\int \left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} \right) dx} = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

Multiplying by x , the DE becomes

$$(2xy + 4x^4)dx + x^2dy = 0$$

$$\text{Check Exact: } \frac{\partial}{\partial y}(2xy + 4x^4) = 2x \quad \frac{\partial}{\partial x}(x^2) = 2x$$

$$f(x, y) = x^2y + 0.8x^5 + c = 0$$

Example:

$$(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$$

Solution:

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = \frac{4}{y} \quad \mu(y) = e^{-\int 4/y dy} = \frac{1}{y^4}$$

Multiply the DE by $\mu(y)$:

$$(2xe^y + 2x/y + 1/y^3)dx + (x^2e^y - x^2/y^2 - 3x/y^4)dy = 0$$

It is Exact and the general solution:

$$x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} + c = 0$$

Example:

$$(10 - 6y + e^{-3x})dx - 2dy = 0$$

Solution:

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = 3 \quad \mu(x) = e^{\int 3dx} = e^{3x}$$

Multiply the DE by $\mu(x)$:

$$(10e^{3x} - 6ye^{3x} + 1)dx - 2e^{3x}dy = 0$$

It is Exact and the general solution:

$$10e^{3x} - 6ye^{3x} + 3x + c = 0$$

Exercise: Solve the following DEs

i. $y' = \frac{6x^5 - 2x + 1}{\cos y + e^y}$

ii. $xy' - y = \sqrt{x^2 + y^2}$

iii. $(1 + e^x y + x e^x y) dx + (x e^x + 2) dy = 0$

iv. $xy dx + (2x^2 + 3y^2 - 20) dy = 0$

v. $(y(x + y + 1)) dx + (x + 2y) dy = 0$

vi. $(10 - 6y + e^{-3x}) dx - 2 dy = 0$

vii. $(y^2 + xy^3) dx + (5y^2 - xy + y^3 \sin y) dy = 0$

Exercise:- Solve the following DEs

Ex1: $y' = \frac{6x^5 - 2x + 1}{\cos y + e^y}$ *Separation*

$$\int (\cos y + e^y) dy - \int (6x^5 - 2x + 1) dx = 0$$

$$\sin y + e^y - x^6 + x^2 - x = c$$

Ex2: $xy' - y = \sqrt{x^2 + y^2}$... *Homg.*

$$y = ux, y' = xu' + u$$

$$xu' + u - u = \sqrt{1 + u^2} \Rightarrow \frac{du}{\sqrt{1 + u^2}} = \frac{dx}{x} \dots \text{Continue}$$

Ex3: $(1 + e^x y + x e^x y) dx + (x e^x + 2) dy = 0$ *Exact*

$$f(x, y) = x e^x y + 2y + x + c = 0$$

Ex4: $xy dx + (2x^2 + 3y^2 - 20) dy = 0$ *Int. Factor*

$$\mu(y) = y^3 \Rightarrow f(x, y) = 0.5x^2 y^3 + 0.5y^6 - 5y^4 + c = 0$$

Exercise:- Solve the following DEs

$$\text{Ex5: } (y(x + y + 1))dx + (x + 2y)dy = 0$$

$$\mu(x) = e^x$$

$$\text{Ex6: } (y^2 + xy^3)dx + (5y^2 - xy + y^3 \sin y)dy = 0$$

$$\mu(y) = \frac{1}{y^3}$$

Linear ODE:

General Form:

$$y' + p(x)y = q(x)$$

Examples: $y' + e^x y = \sin x$

$$y' + \frac{y}{x} = \ln x,$$

$$\frac{dy}{dx} + 3y = \sqrt{1-x}$$

Integrating factor:

$$\mu(x) = e^{\int p(x) dx}$$

General Solution:

$$y(x) = \frac{1}{\mu(x)} \left(\int \mu(x) q(x) dx + c \right)$$

Example:- Solve $(x^2 + 1)y' + xy = x$

$$y' + \frac{x}{x^2 + 1}y = \frac{x}{x^2 + 1}$$

$$\mu(x) = e^{\int \frac{x}{x^2+1} dx} = e^{0.5 \ln(x^2+1)} = \sqrt{x^2 + 1}$$

$$y = \frac{1}{\sqrt{x^2+1}} \left[\int \sqrt{x^2 + 1} \frac{x}{x^2+1} dx + c \right]$$

$$y = \frac{1}{\sqrt{x^2 + 1}} \left[\int \frac{x}{\sqrt{x^2 + 1}} dx + c \right] = \frac{1}{\sqrt{x^2 + 1}} (\sqrt{x^2 + 1} + c)$$

$$y = 1 + \frac{c}{\sqrt{x^2+1}}$$

Example : Dolve the D.E. : $xy' - 2y = x^3 \cos(4x)$

Solution: $y' - \frac{2}{x}y = x^2 \cos(4x)$

$$\mu(x) = e^{\int \frac{-2}{x} dx} = e^{-2 \ln x} = 1/x^2$$

$$y(x) = x^2 \left(\int \frac{1}{x^2} (x^2 \cos 4x) dx + c \right)$$

Example: Solve the D.E. : $x^2 dy + (2xy - x + 1)dx = 0$

Solution

$$\frac{dy}{dx} + \left(\frac{2xy - x + 1}{x^2} \right) = 0$$
$$\frac{dy}{dx} + \frac{2xy}{x^2} = \frac{x - 1}{x^2}$$

$$y' + \left(\frac{2}{x} \right) y = \frac{x - 1}{x^2}$$

$$\mu(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

$$y(x) = \frac{1}{x^2} \left(\int x^2 \left(\frac{x - 1}{x^2} \right) dx + c \right)$$

Summary: Linear ODE

$$y' + p(x)y = q(x) \quad \text{Linear ODE in } y$$

Calculate the integrating factor:

$$\mu(x) = e^{\int p(x)dx}$$

Then its solution can be obtained by:

$$y(x) = \frac{1}{\mu(x)} \left(\int \mu(x)q(x) dx + c \right)$$

$$\frac{dx}{dy} + p(y)x = q(y) \quad \text{Linear ODE in } x$$

Calculate the integrating factor:

$$\mu(y) = e^{\int p(y)dy}$$

Then its solution can be obtained by:

$$x(y) = \frac{1}{\mu(y)} \left(\int \mu(y)q(y) dy + c \right)$$

Example1: Solve the following ODE

$$(2x + 10y^3)dy + ydx = 0$$

Solution

$$\frac{dy}{dx} + \left(\frac{y}{2x + 10y^3} \right) = 0 \quad \text{OR} \quad \frac{dx}{dy} + \left(\frac{2x + 10y^3}{y} \right) = 0$$

$$\frac{dx}{dy} + \frac{2}{y}x = -10y^2$$

$$\mu(y) = e^{\int \frac{2}{y} dy} = e^{2 \ln y} = y^2$$

$$x(y) = \frac{1}{y^2} \left(\int y^2 (-10y^2) dy + c \right)$$

Example 2: Solve the D.E. : $(1 + y^2)dx + (2xy + y^2 + 1)dy = 0$

Solution: $\frac{dx}{dy} + \frac{2y}{1+y^2}x = -1$

$$\mu(y) = e^{\int \frac{2y}{1+y^2} dy} = e^{\ln(1+y^2)} = (1 + y^2)$$

$$x(y) = \frac{1}{y^2 + 1} \left(\int (y^2 + 1)(-1) dy + c \right)$$

Check Examples:- Solve the following DEs.

i. $(3x^2 + y\cos x)dx + (\sin x - 4y^3)dy = 0$

ii. $(4x + xy^2)dx + (y + x^2y)dy = 0$

iii. $(3xy^2 + 2y)dx + (2x^2y + x)dy = 0$

iv. $(2x^3 + y^3)dx - 3xy^2dy = 0$

v. $y' = \frac{1}{x\cos y + a\sin 2y}$

Check Examples:- Solve the following DEs.

i. $(3x^2 + y\cos x)dx + (\sin x - 4y^3)dy = 0$ (Exact)

ii. $(4x + xy^2)dx + (y + x^2y)dy = 0$ (Sep, Exact)

iii. $(3xy^2 + 2y)dx + (2x^2y + x)dy = 0$ ($\mu(x)$)

iv. $(2x^3 + y^3)dx - 3xy^2dy = 0$ ($\mu(x)$)

v. $y' = \frac{1}{x\cos y + a\sin 2y}$ (Linear in x)

Bernoulli DE (Equations Reduced to Linear)

General Form: $y' + p(x)y = q(x)y^n$

Divide by y^n

$$y^{-n}y' + p(x)y^{1-n} = q(x)$$

Let $z = y^{1-n}$, then $z' = (1-n)y^{-n}y'$

$$\frac{z'}{1-n} + p(x)z = q(x)$$

$$z' + (1-n)p(x)z = (1-n)q(x)$$

Linear in z

..... Continue

Example:- $xy - y' = y^3 e^{-x^2}$

Rewrite the ODE as:

$$y' - xy = -y^3 e^{-x^2}$$

The D.E. is (Ber. in y) Divide by y^3 : $y^{-3}y' - xy^{-2} = -e^{-x^2}$

Let $z = y^{-2}$: $z' = -2y^{-3}y'$

$\frac{z'}{-2} - xz = -e^{-x^2} \Rightarrow z' + 2xz = 2e^{-x^2}$ Linear in z

$$\mu(x) = e^{\int 2x dx} = e^{x^2}$$
$$z = e^{-x^2} \left(\int e^{x^2} 2e^{-x^2} dx + c \right) = e^{-x^2} (2x + c)$$

$$y^2 = \frac{1}{e^{-x^2} (2x + c)}$$

Example: $y \frac{dx}{dy} + x = y^3 \sqrt{x}$

Solution:

$$\frac{dx}{dy} + \frac{x}{y} = y^2 \sqrt{x} \quad \text{Ber. In } x: \quad \text{divide by } \sqrt{x} \quad \rightarrow \quad \frac{1}{\sqrt{x}} \frac{dx}{dy} + \frac{\sqrt{x}}{y}$$

• **Let** $z = \sqrt{x}$ $\frac{dz}{dy} = \frac{1}{2\sqrt{x}} \frac{dx}{dy} \rightarrow \frac{dz}{dy} + \frac{z}{2y} = y^2/2$

$$\mu(y) = e^{\int dy/2y} = \sqrt{y}$$

$$z = \frac{1}{\sqrt{y}} \left(c + \int \frac{y^2 \sqrt{y}}{2} dy \right) \rightarrow \sqrt{xy} = c + \frac{y^3 \sqrt{y}}{7}$$

$$x = \left(\frac{c}{\sqrt{y}} + \frac{y^3}{7} \right)^2$$

Example: Solve the DE

Solution: D.E. is $\frac{d\theta}{dt} - \theta \cos t = \theta^2 \cos t (1 - \sin t)$
 (Per. in θ) divide by θ^2

$$\theta^{-2} \frac{d\theta}{dt} - \theta^{-1} \cos t = \cos t (1 - \sin t)$$

$$\varphi = \theta^{-1} \quad \frac{d\varphi}{dt} = -\theta^{-2} \frac{d\theta}{dt}$$

$$\frac{d\varphi}{dt} + \varphi \cos t = \cos t (1 - \sin t)$$

$$\mu(t) = e^{\int \cos t dt} = e^{\sin t}$$

$$\varphi(t) = e^{-\sin t} \left(c + \int e^{\sin t} (1 - \sin t) \cos t dt \right)$$

$$\theta = e^{\sin t} / (c + e^{\sin t} (\sin t - 2))$$

Examples:-Solve the following ODE

i. $xy - y' = y^3 e^{-x^2}$

ii. $y' = xy^2 + 3xy$

iii. $(6y^2 - xy - y)dx + 2xdy = 0$

iv. $y' - y \tan x = \sec x$

Examples:-Solve the following ODE

i. $xy - y' = y^3 e^{-x^2}$ (Ber. in y)

ii. $\frac{dx}{dy} = x^2 y + 3xy$ (Ber. in y)

iii. $(6y^2 - xy - y)dx + 2xdy = 0$ (Ber. in y)

iv. $y' - y \tan x = \sec x$ (Linear in y)

End of Lect. 3